

Metropolitan University
Department of Computer Science and Engineering
Summer Semester Final Examination – 2024
Program: B.Sc. in CSE; Batch: 57
Course: CSE 237: Microprocessor & Interfacing

Time: 2 hour

Marks: 40

Answer any **four** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

1. An 8086 microprocessor is operating in minimum mode and performing a write operation of 2 bytes of data to a memory location starting at an even address.
 - a) Draw the detailed Bus Cycle diagram for this specific operation. 4
 - b) State the values of *BHE* and *A₀* signals for the scenario above. Also, identify which memory banks are accessed and why. 3
 - c) Explain how the *READY* pin affects the write operation and the overall timing if the memory connected to the 8086 is slower than the microprocessor itself. 3
- ✓ 2. An 8086 microprocessor system has three devices connected to an 8259A PIC. All three devices request interrupts at the same time, but only one interrupt can be executed at a time.
 - a) Write down the differences between maskable and non-maskable interrupts. 2
 - b) Explain how the 8259A PIC will determine which interrupt to send to the 8086 microprocessor first. 3
 - c) Describe the steps from when the 8259A PIC receives those interrupts to when the microprocessor executes the first interrupt. 5
- ✓ 3. In a microprocessor system, choosing the right I/O access method and data transfer technique is key to performance, involving memory-mapped I/O, isolated I/O, and DMA.
 - a) Draw the block diagram of basic I/O system. 2
 - b) Compare both the memory-mapped I/O and isolated I/O methods of accessing I/O devices. 4
 - c) Discuss why DMA is preferred over programmed I/O and interrupt-driven I/O. 4
- ✓ 4. A portion of the IVT is given below:

Address	9Eh	9Fh	A0h	A1h	A2h	A3h	A4h	A5h
Content	1Fh	3Ah	B7h	8Eh	4Ch	D2h	9Ah	E5h

- a) Find out the physical address of the ISR for Type-40 interrupt. 3
- b) Design the block diagram where you want to connect 39 I/O devices with the 8086 microprocessor using the 8259A. 4
- c) What happens to the current program being executed by the 8086 microprocessor when an interrupt is requested? 3

5. a) What additional functional chips are included in 80186?
 b) Distinguish between the 80386DX and 80386SX microprocessors.
 c) Demonstrate the stages of pipelining in the Pentium processor briefly.

MOV AX, 3039h
 MOV BX, 5BA0h
 DEC AX
 SUB BX, AX

- a) Calculate and display the updated flag value after executing each instruction.
 b) Using the OP code and RM/MOD table find out the Machine Code for these instructions:
 i) MOV DX, [BX + DI] ii) MOV [SI + A10h], 5h

OP Code Table

Type	OP Code
Immediate to Register	1011
Immediate to Memory Location	1100011
Register/Memory to/from Register	100010

R/M and MOD Table

R/M \ MOD	MOD			11	
	00	01	10	W = 0	W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL	AX
001	[BX] + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL	CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL	DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL	BX
100	[SI]	[SI] + d8	[SI] + d16	AH	SP
101	[DI]	[DI] + d8	[DI] + d16	CH	BP
110	d16	[BP] + d8	[BP] + d16	DH	SI
111	[BX]	[BX] + d8	[BX] + d16	BH	DI

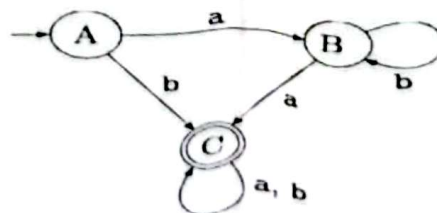
Metropolitan University, Sylhet
 Department of Computer Science and Engineering
 Spring Semester Final Examination – 2024
Program: B.Sc. in CSE; Batch: 57
Course: CSE 327: Theory of Computation

Time: 2 Hours

Marks: 40

Answer any **four sets of questions**. All parts of each question must be answered sequentially.
 Figures in the right margin indicate full marks.

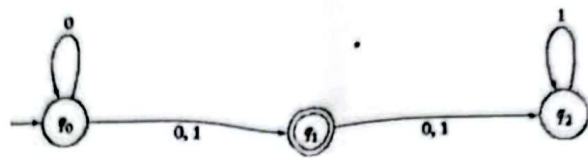
- ✓ 1. a) Find Regular Expressions for the following regular languages: 4 × 1.5 = 6
- $L_1 = \{vwv : w \in \{a, b\}^*, |v| = 2\}$
 - $L_2 = \{\text{Language accepting strings of length at most 2}\}$ where $\Sigma = \{a, b\}$
 - $L_3 = \{\text{All string having even number of 0's}\}$ where $\Sigma = \{0, 1\}$
 - $L_4 = \{\text{Accepting all the strings in which any number of b's follows one or more number of a's is followed by any number of c's}\}$
- b) What are the differences between regular grammar and context-free grammar? 2
- c) Construct right-linear grammars for the following language: 2
- $L = \{a^n b^m : n \geq 2, m \geq 3\}$
- ✓ 2. Consider the grammar with productions $S \rightarrow a \mid Sa \mid bS \mid SSb \mid SbS \mid \lambda$
- Show leftmost derivation, rightmost derivation, and derivation tree for the string "aabbbaaa". 5
 - Find if the given grammar is ambiguous or not. Show the logic behind your claim. 3
 - What are the key properties of parse trees? 2
- ✓ 3. a) Write the formal definition of Grammar. 1
- b) Construct the regular expression for the given FA. 4



- c) Design an FA from the given regular expression $10 + (0 + 11)0^* 1$. 5

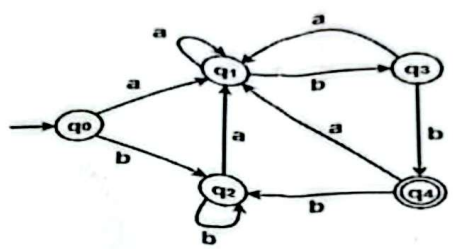
5

a) Convert the following NFA to DFA.



5

b) Reduce the number of states in the following DFA.



5. a) Consider the following rules for a C identifier (c.g. a variable name):
 -an identifier must start with a letter or an underscore
 -an identifier can contain letters, digits, and underscores
 Write a **grammar** that denotes the language containing all valid C identifiers.

5

b) State two fundamental differences between NFA and DFA.

2

c) Determine whether the following grammars are simple. Justify your answer.

3

- i) $G_1 = (\{S, E, F\}, \{+, *, \%, a, b\}, S, P)$ with P given as:
 $S \rightarrow +E \mid *F \mid \%F$, $E \rightarrow \%S \mid \%F \mid \lambda$, $F \rightarrow a \mid b \mid \lambda$
- ii) $G_2 = (\{S, E, F\}, \{+, *, \%, a, b\}, S, P)$ with P given as:
 $S \rightarrow aA \mid bB$, $A \rightarrow aA \mid baA$, $B \rightarrow bb \mid \lambda$

6. a) Write the formal definition of Pushdown Automata.

2

b) Consider the NPDA $M = (\{q_0, q_1, q_2, q_3, q_4\}, \{a, b, c\}, \{0, 1\}, \delta, q_0, 0, \{q_3\})$ where δ is given by

$2 \times 2.5 = 5$

$\delta(q_0, c, 0) = \{(q_0, 0)\}$	$\delta(q_1, a, 1) = \{(q_1, 11)\}$	$\delta(q_2, b, 1) = \{(q_2, \lambda)\}$	$\delta(q_4, c, 0) = \{(q_4, 0)\}$
$\delta(q_0, a, 0) = \{(q_1, 10)\}$	$\delta(q_1, b, 1) = \{(q_2, \lambda)\}$	$\delta(q_2, \lambda, 0) = \{(q_3, \lambda)\}$	$\delta(q_4, \lambda, 0) = \{(q_3, \lambda)\}$
$\delta(q_0, \lambda, 0) = \{(q_1, \lambda)\}$		$\delta(q_2, c, 0) = \{(q_4, 0)\}$	

Draw the NPDA. Also, show if the string **cccaabb** is accepted or not by M.

c) Construct a PDA that can be accepted for the language $L = \{a^n b^{2n} : n \geq 1\}$.

3

Metropolitan University, Sylhet
Department of Computer Science and Engineering
Spring Semester Final Examination – 2024
Program: B.Sc. in CSE; Batch: 57
Course: CSE 223: Database Management System

Time: 2 Hours

Marks: 40

Answer any **four** sets of questions. All parts of each question must be answered sequentially.
Figures in the right margin indicate full marks.

- ✓ 1. a) Insert the following keys in a B+ Tree. 2x4
50 100 49 101 40 102 70 104 106 80 108
I. For $m=3$
II. For $m=4$
- b) Why B+ Tree is used in Database Management System? 2
- ✓ 2. a) What are the rules for a relation to be in BCNF? 2
- b) Find the Candidate Key considering $R=(A, B, C, D, E)$ and $F=\{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$ 5
- c) Find the Canonical Cover of F . 3
- ✓ 3. a) Difference between Inner Join and Outer join. 2
- b) "Candidate key is a subset of a super key". Is this statement True/False? Then explain the reasons with proper examples 4
- c) Explain Set-Intersection operation using Set-Difference operation with a diagram. Give some examples of SQL Aggregate functions. 4
- ✓ 4. Customer(cus_no, cus_name, address, telephone, e-mail)
Order(order_no, order_date, cus_no, total_price, delivery_date)
Order line(order_no, prod_no, quantity, line_total, color, price, unit_price)
Product(prod_no, prod_name, prod_type, unit_price, weight, quantity)
Supplier(sup_no, sup_name, prod_no, telephone, city)
Shipment(prod_no, sup_no, quantity, date, total)
- a) Write SQL: 2x2.5
- I. Find product name for products that either weight more than 50kg or are supplied by supplier SS3.
- II. Find supplier numbers and names for suppliers who do not supply product type sofa or chair.

b) Write relational algebra:

2x2.5

- I. Find all product types which are supplied in the Sylhet city.
- II. Find all customer names whose product's delivery date is '2024-06-23'

5. a) Define ER Diagram and explain the ACID properties of a Transaction.

1+3

b) Check whether the given schedule S is conflict serializable or not. If, yes, then determine all the possible serialized schedule-

T1	T2	T3	T4
			R(A)
	R(A)		
		R(A)	
W(B)			
	W(A)		
		R(P)	
	W(B)		

6. a) Define Canonical Cover and Extraneous Attribute.

2

b) Consider that, $R = (ID, Name, Age)$ $F = \{ID \rightarrow Name, Name \rightarrow Age\}$

8

- I. What are the conditions of 1 NF.
- II. Check whether R is in BCNF or not.
- III. If not mention the reason and
- IV. Decompose R in BCNF

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Course: CSE 237: Microprocessor & Interfacing

Time: 1 hour

Marks: 15

Answer two questions out of the first three. It is mandatory to answer **Question no 4**. Figures in the right margin indicate full marks.

✓

MOV AX, 3C59h
MOV BX, 28FEh
ADD AX, BX

You are learning about microprocessors and assembly language. You wrote a short program with three lines of code. These lines tell the 8086 microprocessor to move data and perform addition.

Explain, step-by-step, how the 8086 microprocessor will execute each of these lines of code. 5

- ✓ 2. a) Describe the key components of a Central Processing Unit and briefly explain the function of each. 2
- b) Elaborate on the fetching cycle, including all the necessary steps. 3
3. a) What is the role of the address bus in accessing memory storage? If a computer has 16GB of RAM, how many bits wide is the address bus? 2
- b) Can multiple logical addresses point to the same physical address? - Explain why or why not. 3

✓

MOV DX, 3C27h
MOV CX, ACBDh
ADD CX, DX

- a) Calculate and display the updated flag value after executing the Add instruction. 2
- b) Using the OP Code and RM/MOD table find out the Machine Code for instructions. 3

MOV [BX] [DI] Ch, E82Ch

OP Code Table

Type	OP Code
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Immediate to Memory Location	1100011
Register/Memory to/from Register	100010

R/M and MOD Table

R/M \ MOD	00	01	10	11	
				W = 0	W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL	AX
001	[BX] + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL	CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL	DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL	BX
100	[SI]	[SI] + d8	[SI] + d16	AH	SP
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Program: B.Sc. in CSE; CSE(DAY)
Batch: 57
Course: CSE 327: Theory of Computation

Time: 1.5 Hours

Marks: 30

Answer any **three** questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

- ✓ 1. Find the grammar and write down the quadruple form. 4*2.5
- a) Given $\Sigma = \{a\}$, $L_1 = \{w: |w| \% 3 \neq 0\}$
 - b) Given $\Sigma = \{a, b\}$, $L_2 = \{a^{2n+1}b^n : n \geq 0\}$
 - c) Given $\Sigma = \{a, b, c\}$, $L_3 = \{w: w \text{ contains the prefix 'aab'}\}$
 - d) Given $\Sigma = \{a, b\}$, $L_4 = \{a^n b^m : n \geq 3, m \geq 2\}$
2. Find a dfa and write down the quintuple form. 4*2.5
- a) Given $\Sigma = \{a, b\}$, $L_5 = \{w: |w| \% 5 \leq 2\}$
 - b) Given $\Sigma = \{a, b\}$, $L_6 = \{ab^na^m : n \geq 2, m \geq 3\}$
 - c) Given $\Sigma = \{a, b\}$, $L_7 = \{w: w \text{ contains the sets of all strings with exactly one } b\}$
 - d) Given $\Sigma = \{a, b\}$, $L_8 = \{w: w \text{ contains the sets of all strings with at least three } a\text{'s}\}$
3. a) Give a formal definition of grammar. 2
- b) Mention two fundamental differences between NFA and DFA. 2
- c) What is an empty string? When do we declare two automata equivalent? 1+2
- d) Which of the strings **aaab**, **abaab**, **aaaabba**, **bbaab** are accepted by the following 3

dfa:

$M = (\{q_0, q_1, q_2\}, \{a, b\}, \delta, q_0, \{q_1\})$ where δ is given by

$\delta(q_0, a) = q_0, \quad \delta(q_0, b) = q_1,$

$\delta(q_1, a) = q_0, \quad \delta(q_1, b) = q_2,$

$\delta(q_2, a) = q_2, \quad \delta(q_2, b) = q_1.$

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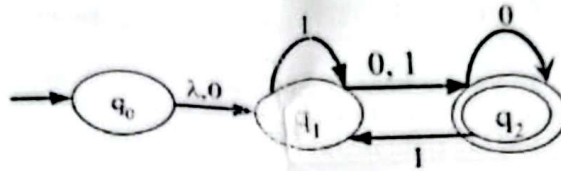
Marks: 30

Answer any **three** questions including question number 2. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.

- ✓ 1. a) Why is DBMS necessary to study in Computer Science and Engineering? 3
- b) Define Primary Key, Super Key, Foreign Key in DBMS. 3
- c) What is the difference between instance and schema. 2
- d) What is DML? Explain it. 2
- ✓ 2. a) *student (rollNo, name, degree, year, gender, deptNo, advisor)* 2 x 5
department (deptNo, name, phone)
professor (empld, name, sex, startYear, deptNo, phone)
course (courseId, cname, credits, deptNo)
enrollment (rollNo, courseId, sem, year, grade)
teaching (empld, courseId, sem, year, classroom)
preRequisite (pr, ReqCourse, courseId)
- Give an expression in SQL for each of the following queries.
- I. Get the rollNo, name of students in the CSE dept (deptNo = 2) along with their advisor's name and phone number
- II. Get the rollNo, name of students who have a lady professor as their advisor
- III. Get the empld, name of the senior most Professor(s)
- IV. Get the classroom and courseId taken by 'Mr Paul' in 2024 'Fall' semester.
- V. Get the name of students along with their department name.
3. a) Define ER-Diagram. Draw an ER-Diagram considering the above schemas with suitable assumptions. (at least four assumption) 2 + 4
eg. A student belongs to only one department
- b) Explain the types of Cardinality in mapping constraints. 4
- ✓ 4. a) Describe 3 levels of Data Abstraction with a diagram. 4
- b) What is the difference between delete and drop command? 1
- c) Define Data Model by giving examples of 3 different data models. 4
- d) Mention the difference between char and varchar data type. 1

✓ a) Convert the following NFA to DFA.

5



b) Reduce the number of states in the following dfa.

5

